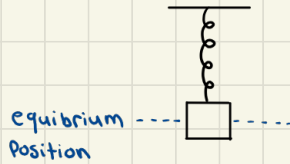


Simple harmonic motion

Consider a mass spring on a spring:



down $\Rightarrow$ Positive direction

up $\Rightarrow$ Negative direction

by hooke's law spring exerts a restoring force F ,

$$\Rightarrow F = -ks$$

↑
Proportional
Constant

We can add the two forces of gravity and spring the spring is going upward so we will have a $-ks$ we will draw a free body diagram to show the forces acting on the system:



so we get:

$$mg - ks = 0$$

• Suppose the mass is in motion we get this DE:

$$m\ddot{x} - \omega^2 x = 0, \quad \omega^2 = \frac{k}{m}$$

when we solve this DE we get:

$$x(t) = C_1 \cos\left(t\sqrt{\frac{k}{m}}\right) + C_2 \sin\left(t\sqrt{\frac{k}{m}}\right)$$

$$\text{cycles} = \frac{bt}{2\pi}$$

Since sine and cosine functions oscillate periodic:

$$f = \frac{b}{2\pi}, \quad P = \frac{2\pi}{b} \quad \text{or} \quad P = \frac{2\pi}{\sqrt{\frac{k}{m}}} \quad b = \omega = \sqrt{\frac{k}{m}}$$

↑ ↑
Frequency Period

— alternate form

$$x(t) = A \sin\left(t\sqrt{\frac{k}{m}} + \phi\right) \quad \leftarrow \begin{array}{l} \text{Phase Shift} \\ \text{this equation and } C_1 \cos(t\sqrt{\frac{k}{m}}) + C_2 \sin(t\sqrt{\frac{k}{m}}) \\ \text{are describing the same motion} \end{array}$$

$$A = \sqrt{C_1^2 + C_2^2}$$

↑
amplitude

$$\begin{aligned} \text{and: } \phi &\Rightarrow \sin(\phi) = \frac{C_2}{A} \\ &\Rightarrow \cos(\phi) = \frac{C_1}{A} \\ &\Rightarrow \tan(\phi) = \frac{C_2}{C_1} \end{aligned}$$

before we continue we need to know what damped and undamped mean:

- undamped motion:

- motion where no energy is lost to the environment (like friction or air resistance)

- key characteristics

- oscillations continue forever
- amplitude is constant
- only the spring constant k and mass m determine motion

- equation of motion:

$$x(t) = A \cos\left(t\sqrt{\frac{k}{m}} + \phi\right) \quad \text{or} \quad x(t) = C_1 \cos\left(t\sqrt{\frac{k}{m}}\right) + C_2 \sin\left(t\sqrt{\frac{k}{m}}\right)$$

• Damped motion:

- motion where energy is gradually lost to resistive forces (friction, air drag)
- Key characteristics
 - amplitude decreases over time (the oscillations "die out" eventually)

— there are different types of damping

- 1) **under damped**: oscillates but the amplitude gradually decreases
- 2) **critically damped**: returns to equilibrium as fast as possible without oscillating
- 3) **over damped**: returns to equilibrium slowly without oscillating

Free damped motion

a spring damped entering a medium, air resistance OR damping device
Velocity slowed by a constant multiple of the velocity called the damping constant B (beta)

$$x'' + \frac{B}{m} x' + \frac{k}{m} x = 0, \quad \frac{B}{m} = 2\lambda, \quad \omega^2 = \frac{k}{m}$$

rewrite

$$x'' + 2\lambda x' + \omega^2 x = 0$$

there are three cases you will most likely encounter:

• Case #1:

$$\lambda^2 - \omega^2 > 0 \quad (\text{over damped})$$

$$x(t) = e^{-\lambda t} \left[C_1 e^{\sqrt{\lambda^2 - \omega^2} t} + C_2 e^{-\sqrt{\lambda^2 - \omega^2} t} \right]$$

over damped: returns to equilibrium slowly without oscillating

• Case #2:

$$\lambda^2 - \omega^2 = 0, \quad (\text{critically damped})$$

$$x(t) = e^{-\lambda t} [C_1 + C_2 t]$$

critically damped: returns to equilibrium as fast as possible without oscillating

• Case #3

$$\lambda^2 - \omega^2 < 0, \quad (\text{under damped})$$

$$x(t) = e^{-\lambda t} \left[C_1 \cos(\sqrt{\omega^2 - \lambda^2} t) + C_2 \sin(\sqrt{\omega^2 - \lambda^2} t) \right]$$

under damped: oscillates but the amplitude gradually decreases

Driven motion

- happens when an external force keeps pushing or pulling the system while it is moving so, instead of the system oscillating on its own, something outside the system is continuously adding energy this external force is called $F(t)$

$$x'' + \frac{B}{m} x' + \frac{k}{m} x = F(t)$$

to actually find $F(t)$ you would need to use this equation:

$$F(t) = \frac{1}{m} F(t)$$


mass

This equation is a non-homogeneous DE so you will have to use techniques from undetermined coefficients and use this equation:

$$x(t) = x_h(t) + x_p(t)$$

you will need this when you are solving problems:

✓ Summary

- Use 9.8 if working in SI units (kg, m, N) — which you are.
- Use 32 if working in Imperial units (lb, ft, lbf).
- The number itself depends entirely on what system of units you are using.

In the English (US) system

- lb (pounds) → weight (force)
- slug → mass

In the SI (metric) system

- kg (kilograms) → mass
- N (newtons) → weight / force

SI and English are very similar

A mass weighing 24 pounds, attached to the end of a spring, stretches it 4 inches. Initially, the mass is released from rest from a point 6 inches above the equilibrium position. (Use $g = 32 \text{ ft/s}^2$ for the acceleration due to gravity.)

Give the initial conditions $x(0)$ (in ft) and $x'(0)$ (in ft/s).

• they want the answer in ft:

$$1 \text{ ft} = 12 \text{ inches}$$

$$S: 4 \text{ inches} \cdot \frac{1 \text{ ft}}{12 \text{ inches}} = \frac{1}{3} \text{ ft}$$

$$S = \frac{1}{3} \text{ ft}$$

$$6 \text{ inches} \cdot \frac{1 \text{ ft}}{12 \text{ inches}} = 0.5$$

$$x(0) = -0.5$$

• above means negative

$$w = 24 \text{ LB}, \quad S = \frac{1}{3} \text{ ft}, \quad x(0) = -0.5, \quad x'(0) = 0$$

• you aren't told anything about velocity

because we are in the imperial / english system the 24 LB is weight

$$w = m \cdot g \rightarrow m = \frac{w}{g} = \frac{24}{32} = 0.7$$

$$k = \frac{m \cdot g}{S} = \frac{(0.7)(32)}{\frac{1}{3}} = 72$$

$$b = \sqrt{\frac{k}{m}} = \sqrt{\frac{72}{0.7}} = 4\sqrt{6}$$

$$x(t) = C_1 \cos\left(t \sqrt{\frac{k}{m}}\right) + C_2 \sin\left(t \sqrt{\frac{k}{m}}\right)$$

$$x(t) = C_1 \cos(4\sqrt{6} t) + C_2 \sin(4\sqrt{6} t)$$

$$x(t) = C_1 \cos(4\sqrt{6} t) + C_2 \sin(4\sqrt{6} t)$$

$$x(0) = -0.5$$

$$x'(t) = -4\sqrt{6} C_1 \sin(4\sqrt{6} t) + 4\sqrt{6} C_2 \cos(4\sqrt{6} t)$$

$$x'(0) = 0$$

$$\begin{cases} C_1 + 0 = -0.5 \\ 0 + 4\sqrt{6} C_2 = 0 \end{cases}$$

★ Plug C_1 and C_2 into $x(t)$

$$x(t) = -\frac{1}{2} \cos(4\sqrt{6} t)$$

$$C_1 = -0.5$$

$$C_2 = 0$$

• notice that this problem is undamped there is no energy that is being lost to environment

A force of 880 newtons stretches a spring 4 meters. A mass of 55 kilograms is attached to the end of the spring and is initially released from the equilibrium position with an upward velocity of 10 m/s.

Give the initial conditions $x(0)$ (in m) and $x'(0)$ (in m/s).

$$F = 880 \text{ N}, \quad S = 4 \text{ m}, \quad m = 55 \text{ kg}, \quad x(0) = 0, \quad x'(0) = -10 \quad \leftarrow \text{negative because its going upward}$$

$$F = kS \Rightarrow k = \frac{F}{S} \Rightarrow \frac{880}{4} = k = 220$$

$$b = \sqrt{\frac{k}{m}} = \sqrt{\frac{220}{55}} = b = 2$$

$$x(t) = C_1 \cos(t \sqrt{\frac{k}{m}}) + C_2 \sin(t \sqrt{\frac{k}{m}})$$

$$x(t) = C_1 \cos(2t) + C_2 \sin(2t)$$

$$x(t) = C_1 \cos(2t) + C_2 \sin(2t)$$

$$x(0) = 0$$

$$x'(t) = -2C_1 \sin(2t) + 2C_2 \cos(2t)$$

$$x'(0) = -10$$

$$\begin{cases} C_1 + 0 = 0 \\ 0 + 2C_2 = -10 \end{cases}$$

★ Plug C_1 and C_2 into $x(t)$

$$x(t) = -5 \sin(2t)$$

$$C_1 = 0$$

$$C_2 = -5$$

A mass weighing 32 pounds stretches a spring 2 feet. Determine the amplitude (in feet) and period of motion (in seconds) if the mass is initially released from a point 1 foot above the equilibrium position with an upward velocity of 6 ft/s. (Use $g = 32 \text{ ft/s}^2$ for the acceleration due to gravity.)

how many complete cycles will the mass have completed by the end of 6π seconds?

you are given 32 LB so that will be weight not force

$$w = 32 \text{ LB}, \quad \Delta = 2 \text{ ft}, \quad x(0) = -1, \quad x'(0) = 6$$

$$w = mg \Rightarrow m = \frac{w}{g} = \frac{32}{32} = m = 1$$

$$k = \frac{mg}{\Delta} = \frac{32}{2} = k = 16, \quad b = \sqrt{\frac{k}{m}} = \sqrt{\frac{16}{1}} = b = 4$$

$$x(t) = C_1 \cos(4t) + C_2 \sin(4t)$$

$$x(0) = -1$$

$$x'(t) = -4C_1 \sin(4t) + 4C_2 \cos(4t)$$

$$x'(0) = 6$$

$$\begin{aligned} x(0) = -1 &\rightarrow \begin{cases} C_1 + 0 = -1 \\ 0 + 4C_2 = 6 \end{cases} = \begin{aligned} C_1 &= -1 \\ C_2 &= 3/2 \end{aligned} \\ x'(0) = 6 &\rightarrow \end{aligned}$$

$$A = \sqrt{C_1^2 + C_2^2} \Rightarrow \sqrt{(-1)^2 + (-3/2)^2} \Rightarrow \boxed{A = \frac{\sqrt{13}}{2}}$$

$$P = \frac{2\pi}{b} \Rightarrow \frac{2\pi}{4} = \boxed{P = \pi/2}$$

$$\text{cycles} = \frac{bt}{2\pi} = \frac{4(6\pi)}{2\pi} = \boxed{12 \text{ cycles}}$$

we know that $t=0$ because we are told "initially released" we also know that $x(0)=1$ and $x'(0)=-6$ because we are told you can think of these two as:

$x(0)=1 \rightarrow$ displacement at the start
 $x'(0)=-6 \rightarrow$ velocity at the start

• Keep in mind both are at $t=0$

we also know that this problem is undamped because we aren't told about energy loss to environment

A 1-kilogram mass is attached to a spring whose constant is 14 N/m, and the entire system is then submerged in a liquid that imparts a damping force numerically equal to 9 times the instantaneous velocity. Determine the equations of motion (in m) if the following situations occur.

(a) the mass is initially released from rest from a point 1 meter below the equilibrium position

$$x(t) = \boxed{1.4e^{-2t} - 0.4e^{-7t}} \quad \checkmark \quad \text{m}$$

(b) the mass is initially released from a point 1 meter below the equilibrium position with an upward velocity of 13 m/s

$$x(t) = \boxed{-1.2e^{-2t} + 2.2e^{-7t}} \quad \checkmark \quad \text{m}$$

we are told that we have 1kg which is the mass

$$m = 1\text{kg}, \quad k = 14 \text{ N/m}, \quad B = 9, \quad x(0) = 1, \quad x'(0) = 0$$

$$x'' + \frac{B}{m}x' + \frac{k}{m}x = 0, \quad \frac{B}{m} = 2\lambda, \quad \omega^2 = \frac{k}{m}$$

↓

$$x'' + 2\lambda x' + \omega^2 x = 0$$

$$2\lambda = 9$$

$$\omega^2 = 14$$

$$x'' + 9x' + 14x = 0$$

• you can now rewrite this in terms of any letter and apply quadratic formula

$$r^2 + 9r + 14 = 0$$

$$a = 1$$

$$b = 9$$

$$c = 14$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\rightarrow \frac{- (9) \pm \sqrt{(9)^2 - 4(1)(14)}}{2(1)}$$

$$\rightarrow \frac{-9 \pm \sqrt{25}}{2}$$

$$r = -2,$$

$$r = -7$$

we already know this problem is damped because it told us "damping force numerically equal to 9 time" but we still need to figure out what equation it is from the 3 damping equation if we look at the **discriminate** we notice it's greater the zero so this will be overdamped or case #1

$$x(t) = e^{-\lambda t} \left[c_1 e^{\sqrt{\lambda^2 - \omega^2} t} + c_2 e^{-\sqrt{\lambda^2 - \omega^2} t} \right]$$

$$2\lambda = 9$$

$$\lambda = 4.5$$

$$\omega^2 = 14$$

$$\omega = \sqrt{14}$$

$$\sqrt{(4.5)^2 - (\sqrt{14})^2} = 2.5$$

$$x(t) = e^{-4.5t} \left[c_1 e^{2.5t} + c_2 e^{-2.5t} \right]$$

$$x(t) = c_1 e^{-2t} + c_2 e^{-7t}$$

$$x(0) = 1$$

$$x'(t) = -2c_1 e^{-2t} - 7c_2 e^{-7t}$$

$$x'(0) = 0$$

$$\begin{aligned} x(0) = 1 &\rightarrow \begin{cases} c_1 + c_2 = 1 \\ -2c_1 - 7c_2 = 0 \end{cases} \rightarrow c_1 = 1 - c_2 \\ x'(0) = 0 &\rightarrow \end{aligned}$$

$$-2(1 - c_2) - 7c_2 = 0$$

$$-2 + 2c_2 - 7c_2 = 0$$

$$c_2 = -0.4$$

$$c_1 = 1 - c_2$$

$$c_1 = 1 - (-0.4)$$

$$c_1 = 1.4$$

a)

$$x(t) = 1.4e^{-2t} - 0.4e^{-7t}$$

b)

We already know the equation is: $x(t) = c_1 e^{-2t} + c_2 e^{-7t}$ so all we need to do is solve for c_1 and c_2 so:

$$x(0) = 1, \quad x'(0) = -13$$

$$x(t) = c_1 e^{-2t} + c_2 e^{-7t}$$

$$x'(t) = -2c_1 e^{-2t} - 7c_2 e^{-7t}$$

$$\begin{aligned} x(0) = 1 &\rightarrow c_1 + c_2 = 1 \rightarrow c_1 = 1 - c_2 \\ x'(0) = -13 &\rightarrow -2c_1 - 7c_2 = -13 \end{aligned}$$

$$-2(1 - c_2) - 7c_2 = -13$$

$$-2 + 2c_2 - 7c_2 = -13$$

$$-2 - 5c_2 = -13$$

$$-5c_2 = -11$$

$$c_2 = \frac{11}{5}$$

$$c_2 = 2.2$$

$$c_1 = 1 - c_2$$

$$c_1 = 1 - 2.2$$

$$c_1 = -1.2$$

$$x(t) = -1.2e^{-2t} + 2.2e^{-7t}$$

• the velocity is negative because it is going upward

A mass of 1 slug, when attached to a spring, stretches it 2 feet and then comes to rest in the equilibrium position. Starting at $t = 0$, an external force equal to $f(t) = 6 \sin(4t)$ is applied to the system. Find the equation of motion (in ft) if the surrounding medium offers a damping force that is numerically equal to 8 times the instantaneous velocity. (Use $g = 32 \text{ ft/s}^2$ for the acceleration due to gravity.)

$$m = 1 \text{ slug}, \quad \delta = 2 \text{ ft}, \quad B = 8, \quad x(0) = 0, \quad x'(0) = 0$$

• it says that it comes to rest in a equilibrium position

$$k = \frac{mg}{\delta} = \frac{(1)(32)}{2} \Rightarrow k = 16, \quad f(t) = 6 \sin(4t)$$

$$x'' + \frac{B}{m}x' + \frac{k}{m}x = f(x), \quad \frac{B}{m} = \frac{8}{1} = 8, \quad \frac{k}{m} = \frac{16}{1} = 16$$

$$f(x) = \frac{1}{m} \cdot f(t) \Rightarrow \frac{1}{1} \cdot 6 \sin(4t) \Rightarrow 6 \sin(4t)$$

$$x'' + 8x' + 16x = 6 \sin(4t)$$

• this is a non-homogeneous equation

$$r^2 + 8r + 16 = 0$$

$$(r + 4)^2 = 0$$

$$r = -4, -4$$

• repeating

$$x_g(t) = C_1 e^{-4t} + C_2 t e^{-4t}$$

• because this is a non-homogeneous we will need to apply undetermined coefficient

$$x_p(t) = C_3 \sin(4t) + C_4 \cos(4t)$$

$$x_p'(t) = 4C_3 \cos(4t) - 4C_4 \sin(4t)$$

$$x_p''(t) = -16C_3 \sin(4t) - 16C_4 \cos(4t)$$

$$(-16C_3 \sin(4t) - 16C_4 \cos(4t)) + 8(4C_3 \cos(4t) - 4C_4 \sin(4t)) + 16(C_3 \sin(4t) + C_4 \cos(4t)) = 6 \sin(4t)$$

$$-16C_3 \sin(4t) - 16C_4 \cos(4t) + 32C_3 \cos(4t) - 32C_4 \sin(4t) + 16C_3 \sin(4t) + 16C_4 \cos(4t) = 6 \sin(4t)$$

$$3(c_3 \cos(4t)) = 0$$

$$-3(2c_4 \sin(4t)) = 6 \sin(4t)$$

$$c_3 = 0$$

$$c_4 = -\frac{3}{16}$$

$$x_p(t) = c_3 \sin(4t) + c_4 \cos(4t)$$

$$x_p(t) = -\frac{3}{16} \cos(4t)$$

$$x(t) = x_g + x_p \Rightarrow x(t) = c_1 e^{-4t} + c_2 t e^{-4t} - \frac{3}{16} \cos(4t)$$

$$x(t) = c_1 e^{-4t} + c_2 t e^{-4t} - \frac{3}{16} \cos(4t)$$

$$x'(t) = -4c_1 e^{-4t} + c_2 e^{-4t} - 4c_2 t e^{-4t} + \frac{3}{4} \sin(4t)$$

$$\begin{cases} x(0) = 0 & c_1 + 0 - \frac{3}{16} = 0 \\ x'(0) = 0 & -4c_1 + c_2 - 0 + 0 = 0 \end{cases} \Rightarrow c_1 = 3/16$$

$$-4(3/16) + c_2 = 0$$

$$c_2 = 3/4$$

$$x(t) = \frac{3}{16} e^{-4t} + \frac{3}{4} t e^{-4t} - \frac{3}{16} \cos(4t)$$

this problem was a driven motion because we were told "an external force equal to $F(t) = 6 \sin(4t)$ is applied to the system" which resulted in a non-homogeneous equation we had to solve using undetermined coefficients